Complexity guarantees and polling strategies for Riemannian direct-search methods

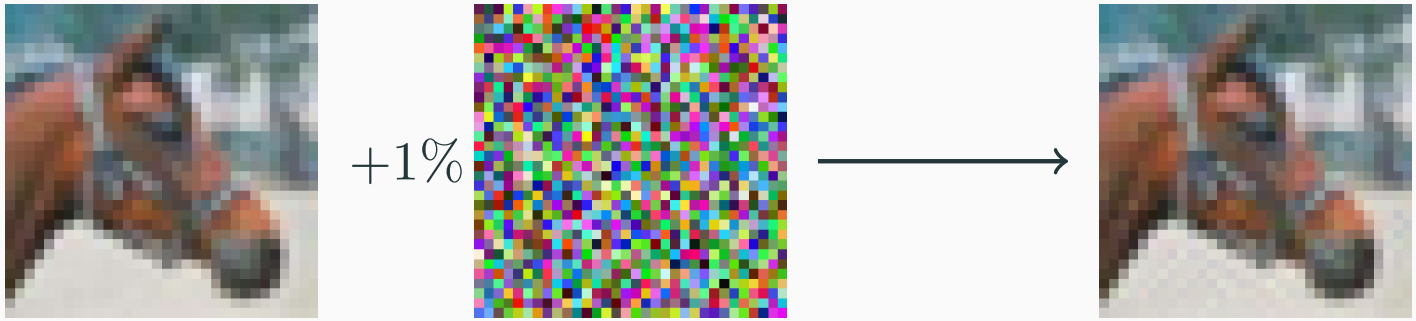
Bastien Cavarretta, *F. Goyens, C. W. Royer and F. Yger*

JKU Linz | 9. Juli 2026





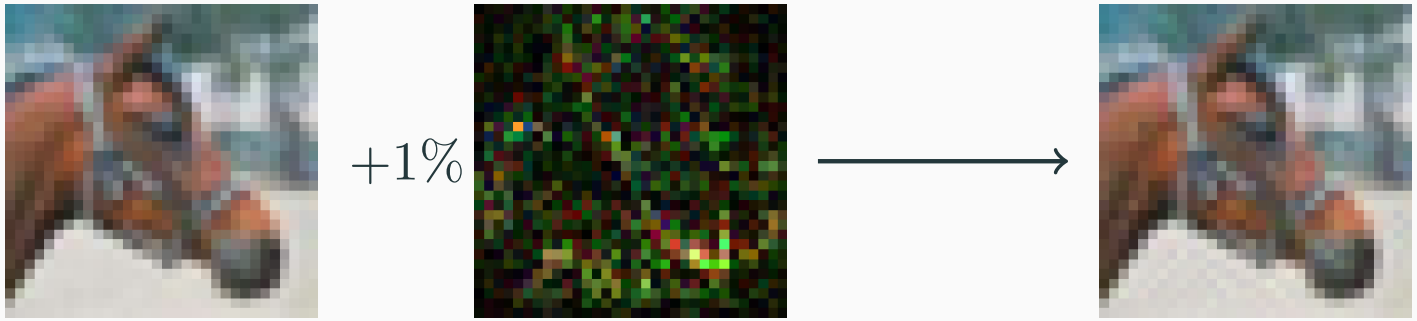
Motivation: Adversarial attacks of Neural Networks



"horse" perturbation 1024 pixels "horse"

Goal: Fool a classifier g : images $\rightarrow$ label.

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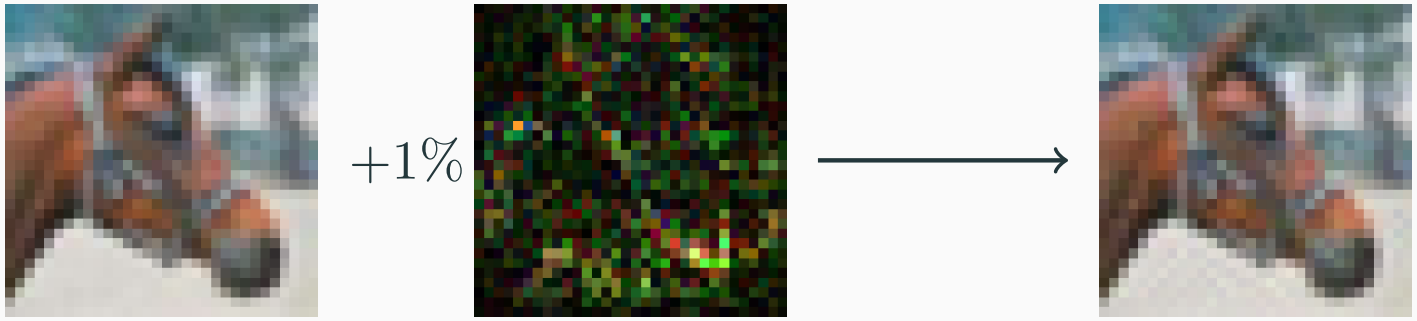


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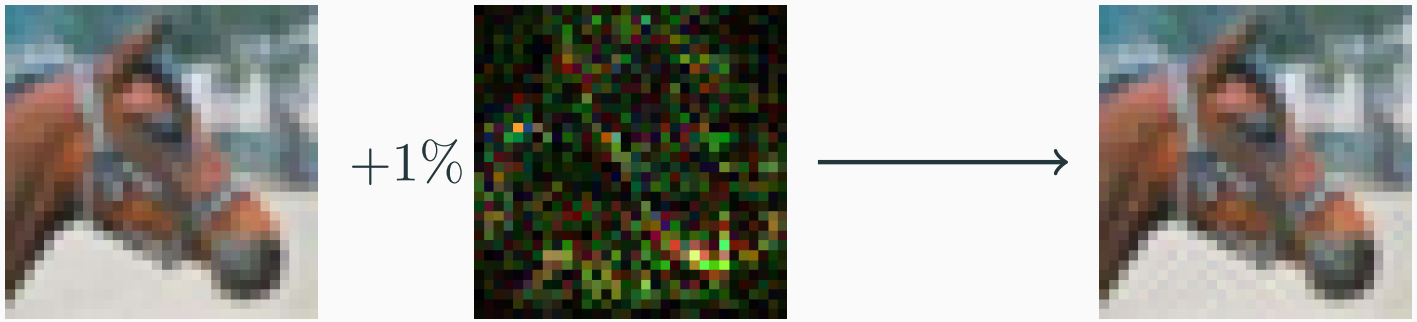
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$$\max_{\text{small}} \quad \text{ce}(g(\text{horse} + \text{noise}), \text{”horse”})$$

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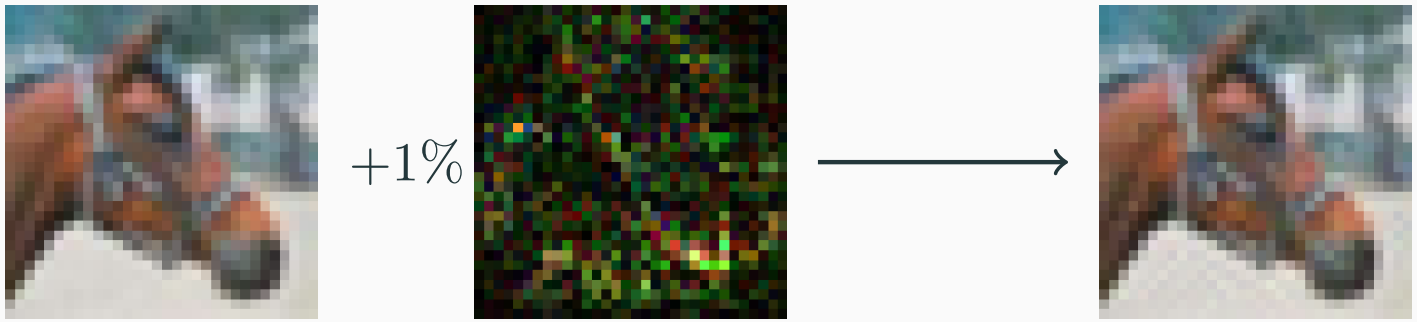
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$$\max_{\substack{\delta \in \mathbb{R}^{1024 \times 3} \\ \|\delta\|_2 \leq 0.5}} \text{ce}(g(x+\delta), y)$$

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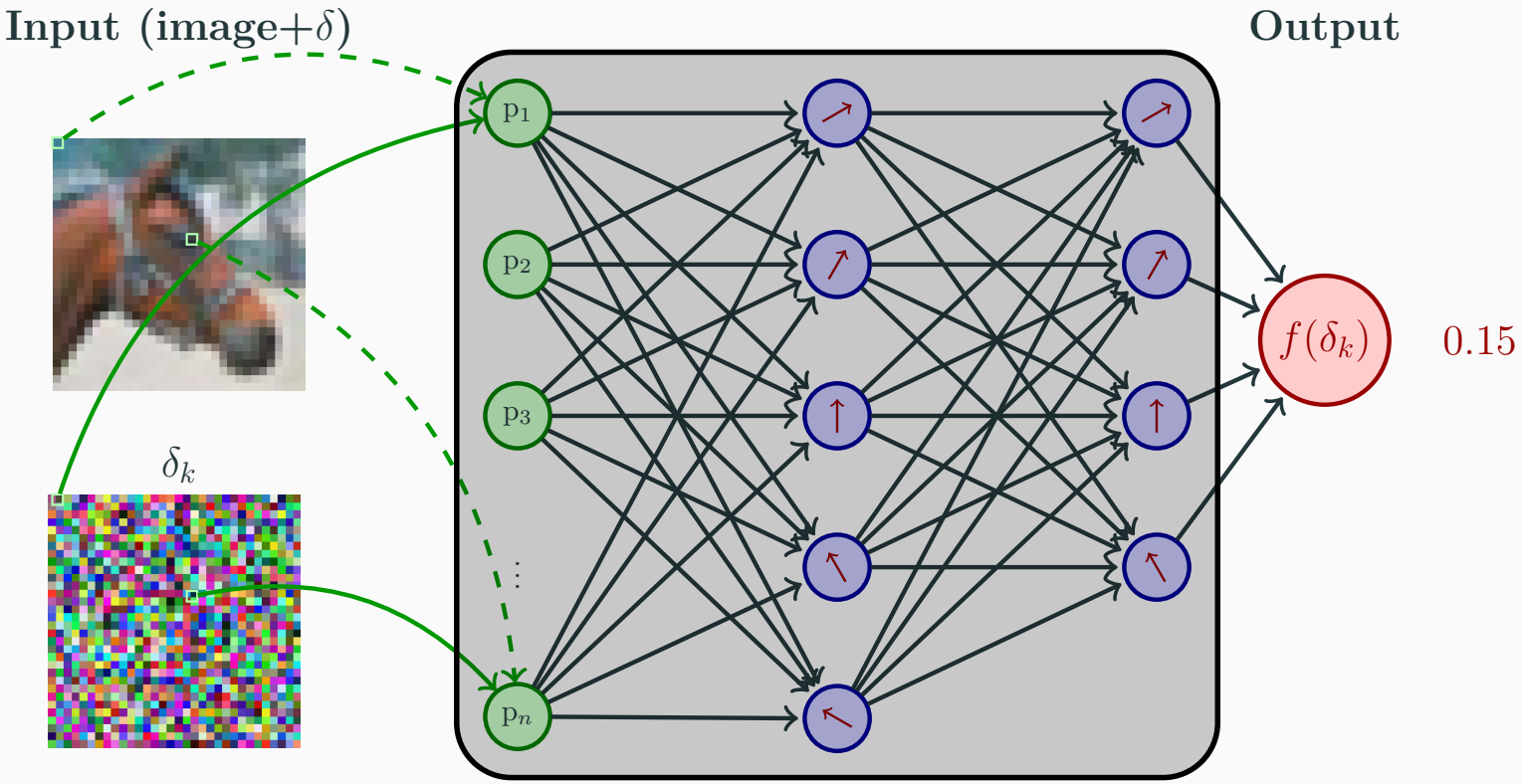
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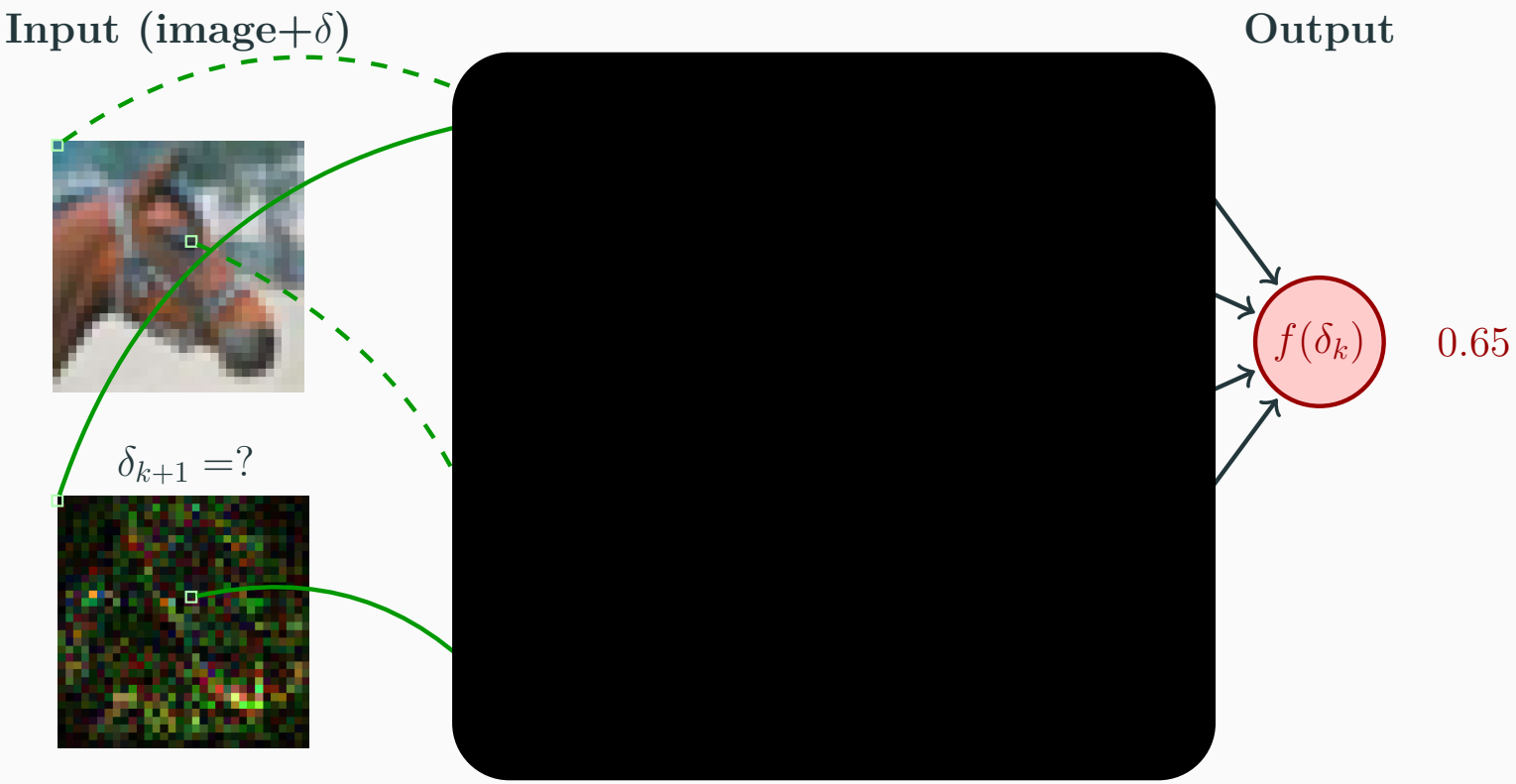
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Black-box attacks (1/2)



Whitebox attack (gradients ✓): $\delta_{k+1} = \delta_k + \gamma \nabla_\delta f(\delta_k)$

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Black-box attack (gradients ✗): $\delta_{k+1} = ?$

$\rightarrow$ **Derivative-free optimization**

Black-box attacks (2/2)

Need for scalable methods

- CIFAR10 : dimension 1024×3
- Subspace approaches (Chen et al., ZOO, '17)
- More fancy attacks : Wavelet domain (Cai et al. '21). Dimension $> 10^6$.

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Setting

$$\min_{x \in \mathcal{M}} f(x)$$

- $\mathcal{M} \subset \mathbb{R}^n$ Riemannian manifold. $\dim(\mathcal{M}) = m \leq n$.
- f smooth, black-box, bounded below.

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Our work

- Complexity bound for direct search
- Guidelines for implementing polling directions

Table of contents

Riemannian Direct Search and Complexity

Projected or intrinsic polling

Numerical experiments

Riemannian Direct Search and Complexity

Riemannian direct-search

Input $x_0 \in \mathbb{R}^m$, $0 < \alpha_0 < \alpha_{\max}$, $c > 0$.

For $k = 1, 2, \ldots$

- Take a PSS $D \subset \mathbb{R}^m$.
- If there exists $d \in D_k$ s.t. $f(x_k + \alpha_k d) \leq f(x_k) - c\alpha_k^2 \|d\|^2$,
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- Else, $x_{k+1} = x_k, \ \alpha_{k+1} = \frac{1}{2}\alpha_k$.

Kungurtsev, Rinaldi, Zeffiro, *Retraction based Direct Search Methods for Derivative Free Riemannian Optimization*, JOTA, 2024

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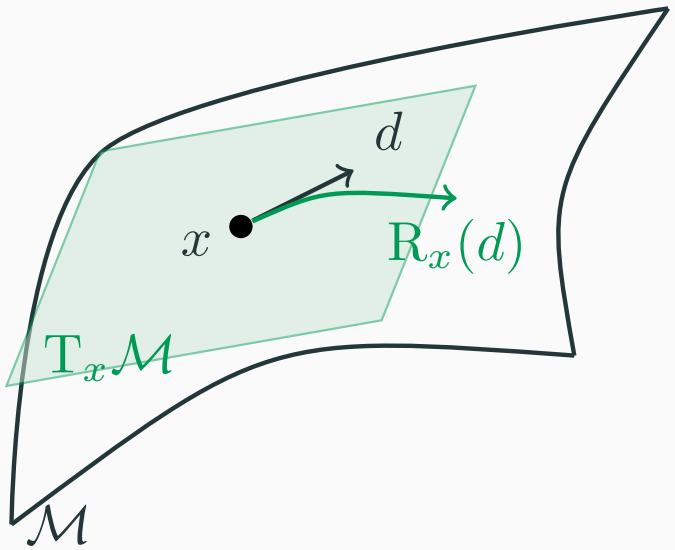
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What changes

$\mathrm{T}_x\mathcal{M}$: Tangent space of $\mathcal{M}$ at x.
$\mathrm{R}_x(d)$: Retraction, move on the manifold with initial speed d.



Kungurtsev, Rinaldi, Zeffiro, *Retraction based Direct Search Methods for Derivative Free Riemannian Optimization*, JOTA, 2024

Convergence of Riemannian direct-search

Asymptotic convergence (Kungurtsev, Rinaldi, Zeffiro '24)

If $\mathrm{grad} f$ is L-Lipschitz and directions D_k are well-chosen, then,

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- D_k computed by projecting PSSs of $\mathbb{R}^n$ into $\mathrm{T}_x\mathcal{M}$.

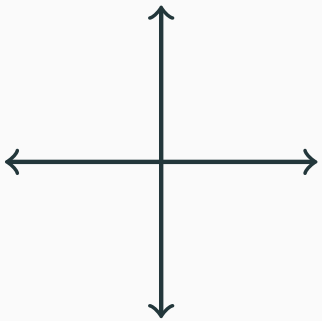
What is a *good set* of directions in $\mathbb{R}^m$?

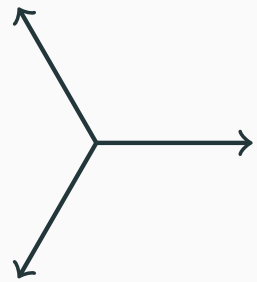
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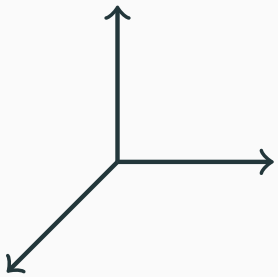
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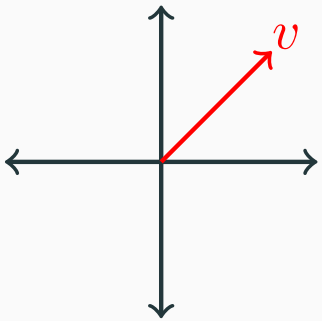
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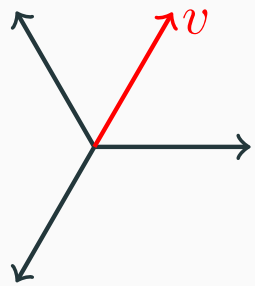
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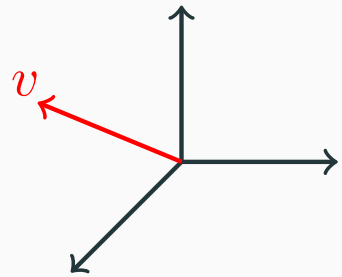
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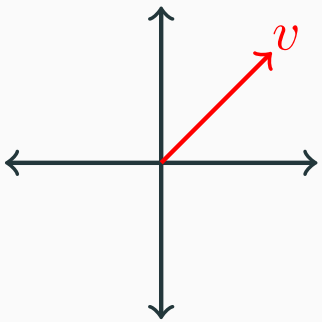
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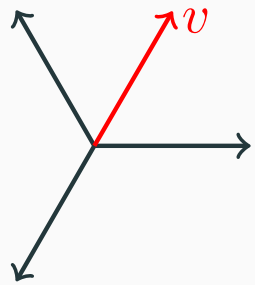
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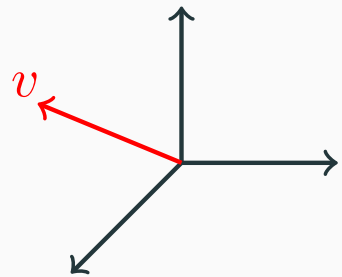
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Context: $\min_{\mathbb{R}^m} f$ with fixed PSS D (Euclidean)

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If ∇f is L-Lipschitz, the method reaches $\|\nabla f(x_k)\| \leq \epsilon$ in less than

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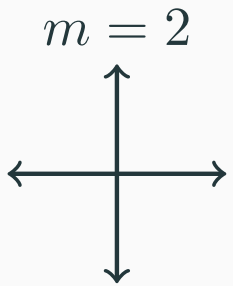
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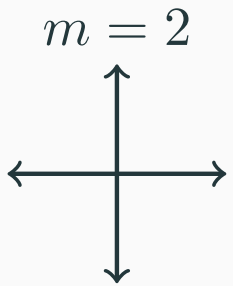
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Generalizing to the whole manifold

Let $\mathcal{D} = (D_x)_{x\in\mathcal{M}}$, $D_x \in \mathrm{T}_x\mathcal{M}$.

Cosine Measure of $\mathcal{D}$:

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$\mathcal{D}$ is a PSS for $\mathcal{M}$ if, $\operatorname{cm}(\mathcal{D}) > 0$.

Riemannian direct-search complexity

Context: $\min_{x \in \mathcal{M}} f$.

Assume:

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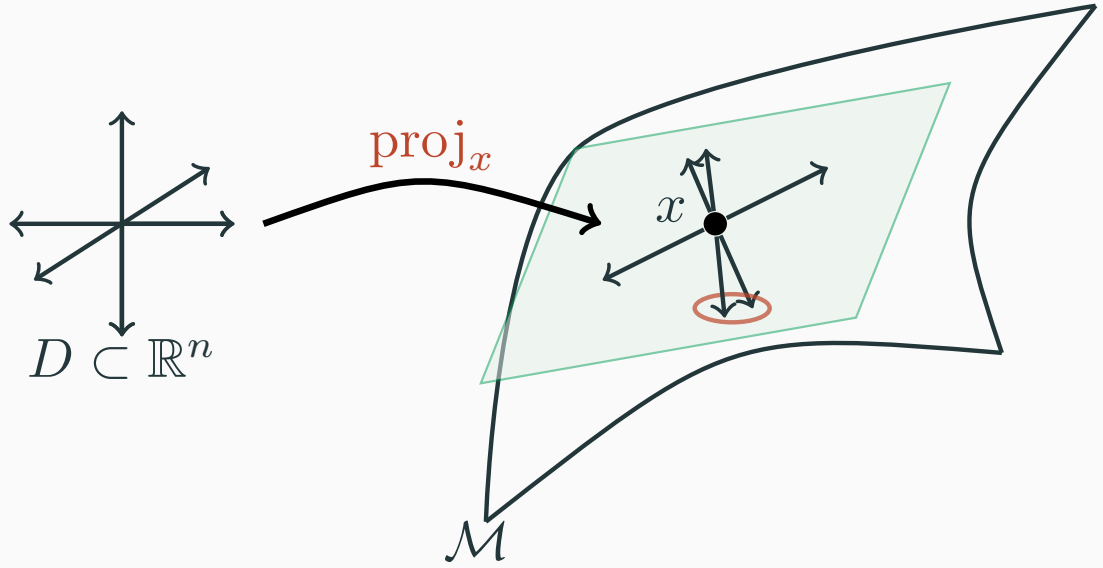
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Projected or intrinsic polling

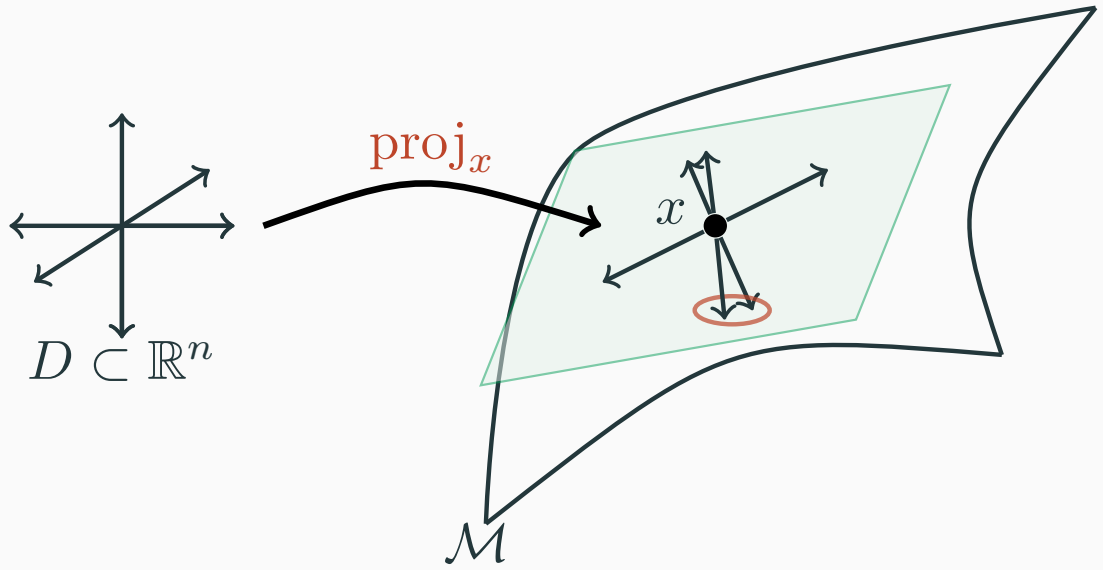
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Idea Project an ambient PSS in $\mathbb{R}^n$ to every $\mathrm{T}_x\mathcal{M}$.



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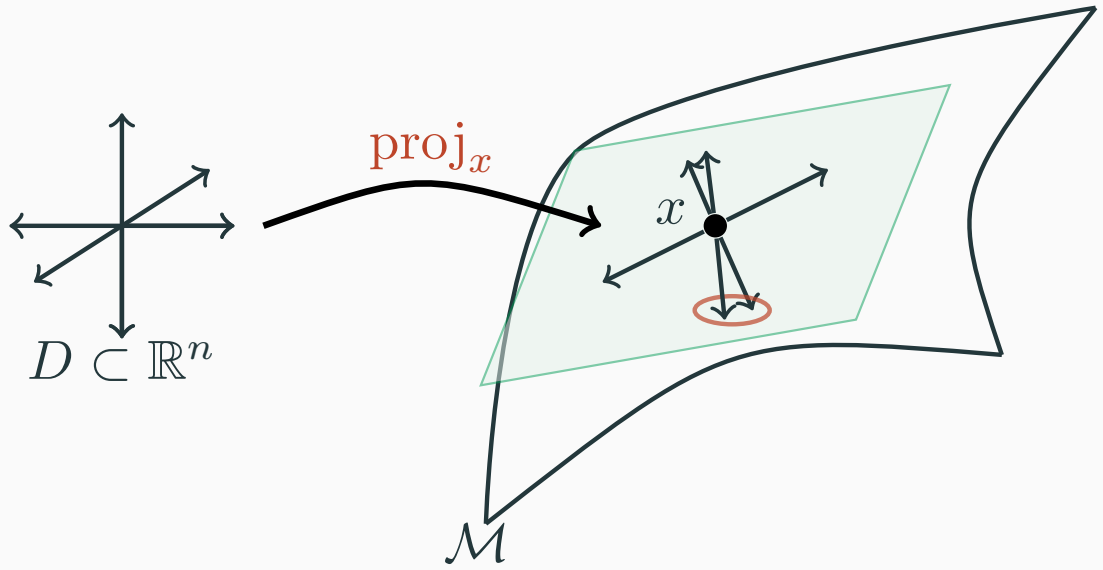


Projected PSS: Denote $P_x(D)$ projection of D onto $\mathrm{T}_x\mathcal{M}$,

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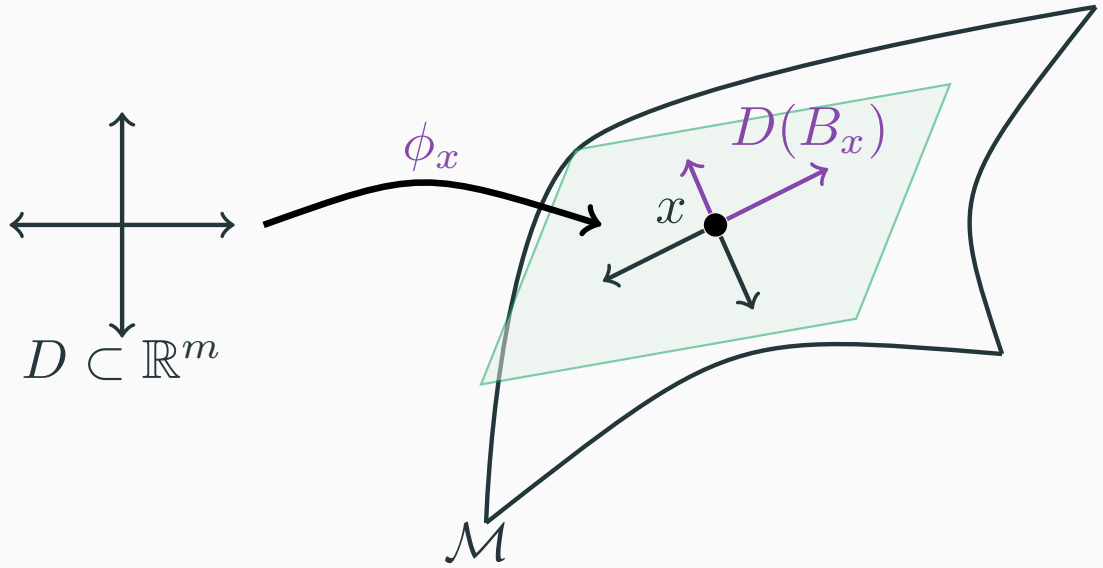
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$$\mathrm{cm}_{\mathcal{M}}(\mathcal{P}(D)) \geq \mathrm{cm}(D),\ \chi_{\mathcal{M}}(\mathcal{P}(D)) \leq \chi_n(D)$$

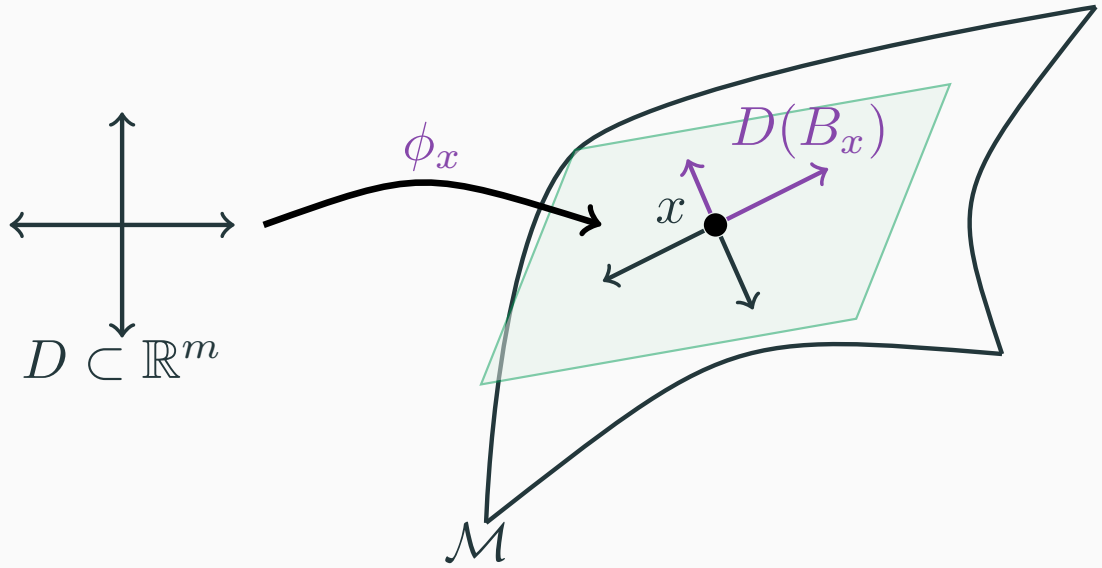
PSS for manifolds: Intrinsic way

Idea Use isomorphism induced by orthogonal bases between $\mathbb{R}^m$ and $\mathrm{T}_x\mathcal{M}$.



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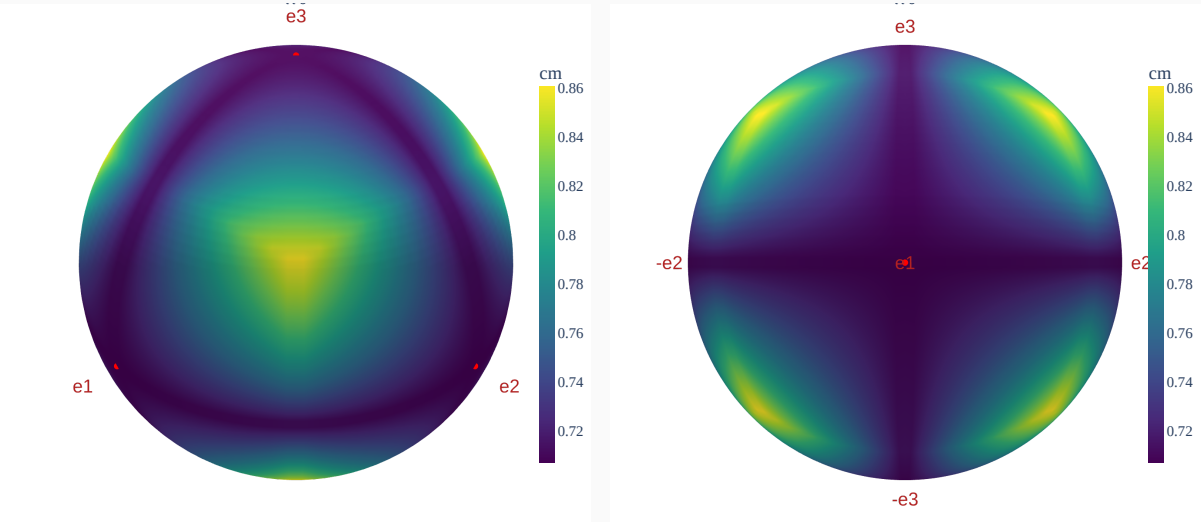


Intrinsic PSS: From D PSS of $\mathbb{R}^m$ and $\mathcal{B} = (B_x)_{x\in\mathcal{M}}$ bases, build

$$D(\mathcal{B}) := (D(B_x))_{x\in\mathcal{M}}.$$

$$\mathrm{cm}_{\mathcal{M}}(D(\mathcal{B})) = \mathrm{cm}(D),\ \chi_{\mathcal{M}}(\mathcal{P}(D)) = \chi_m(D)$$

A simple case: $\mathcal{M} = \mathbb{S}^{n-1} := \{x \in \mathbb{R}^n \mid \|x\|^2 = 1\}$



heatmap of $\mathrm{cm}_{\mathrm{T}_x\mathcal{M}}(P_x(D^{\oplus}))$

	Projected	Intrinsic
PSS	$\mathcal{P}(D^{\oplus})$	$D^{\oplus}(B)$
size ℓ		$2(n-1)$
cm		$\frac{1}{\sqrt{n-1}}$
χ		$2(n-1)^2$

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Theorem (C., Goyens, Royer, Yger '25+)

On $\mathcal{M} = \mathbb{S}^{n-1}$,

$$\frac{1}{\sqrt{n-2+\|x\|_\infty^2}} \geq \mathrm{cm}_x(P_x(D^{\oplus})) \geq \frac{1}{\sqrt{n-1}},$$

and second inequality is tight

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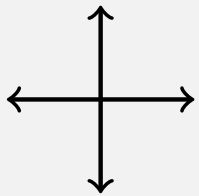
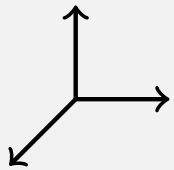
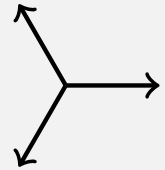
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	Projected	Intrinsic
PSS	$\mathcal{P}(D^{\oplus})$	$D^{\oplus}(B)$
size ℓ	$2n$	$2(n-1)$
cm	$\geq \frac{1}{\sqrt{n-1}}$	$\frac{1}{\sqrt{n-1}}$
χ	$2n(n-1)$	$2(n-1)^2$

Numerical experiments

Positive spanning sets in $\mathbb{R}^m$

PSS	$D^{\oplus}$	$D^{\dbinom{}{}}$	$D^{\circledast}$
			
$\lvert\cdot\rvert$	$2m$	$m+1$	$m+1$
$\mathrm{cm}(\cdot)$	$\frac{1}{\sqrt{m}}$	$\frac{1}{\sqrt{m^2+2(m-1)\sqrt{m}}}$	$\frac{1}{m}$
$\chi(\cdot)$	$2m^2$	$m^3+O(m^{5/2})$	$m^3+O(m^2)$
	$D^{\oplus}$ (intr) $D^{\oplus}$ (proj)	$D^{\dbinom{}{}}$ (intr) $D^{\dbinom{}{}}$ (proj)	$D^{\circledast}$ (intr) $D^{\circledast}$ (proj)

Toy manifold problems

Problems

Manifold	$f(x)$	Parameters
$F := \operatorname{span}(z_1, \ldots, z_m)$	$\sum_{i=1}^{10} \|x_i - x\|_2^2$	$(x_i) \in (\mathbb{R}^n)^{10}$
	$\sum_{i=1}^{10} \|x_i - x\|_2^2$	$(x_i) \in F^{10}$
	$\frac{1}{2}\langle Ax, x\rangle - \langle b, x\rangle$	$A \succcurlyeq \frac{1}{10} I_n,\ b \in \mathbb{R}^n$
$\mathbb{S}^{(m+1)-1} \times \{0_{\mathbb{R}^{n-m}}\}$	$\langle x, Ax\rangle$	$A \in \mathcal{S}ym(n)$

Toy manifold problems

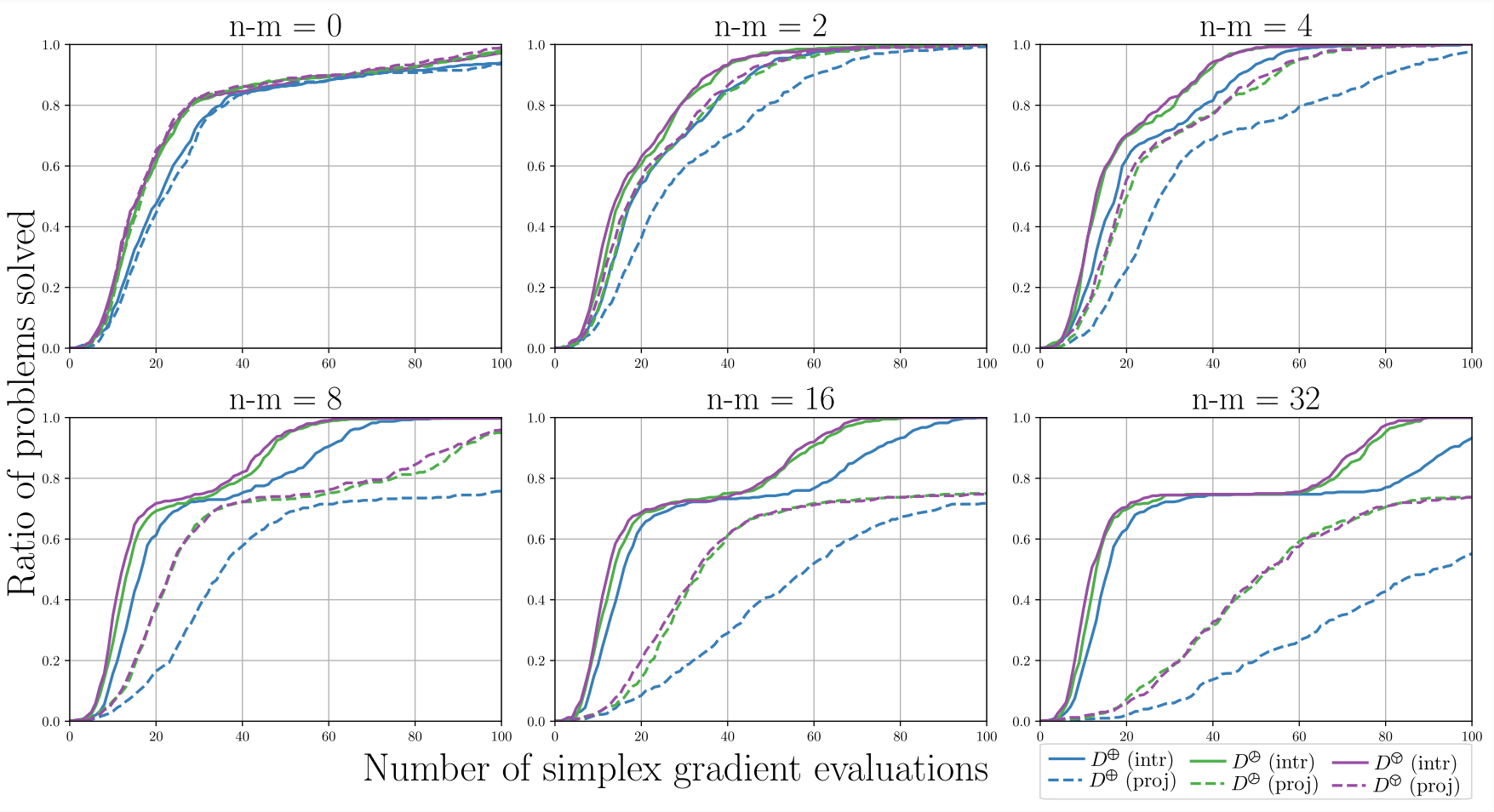
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Data profiles

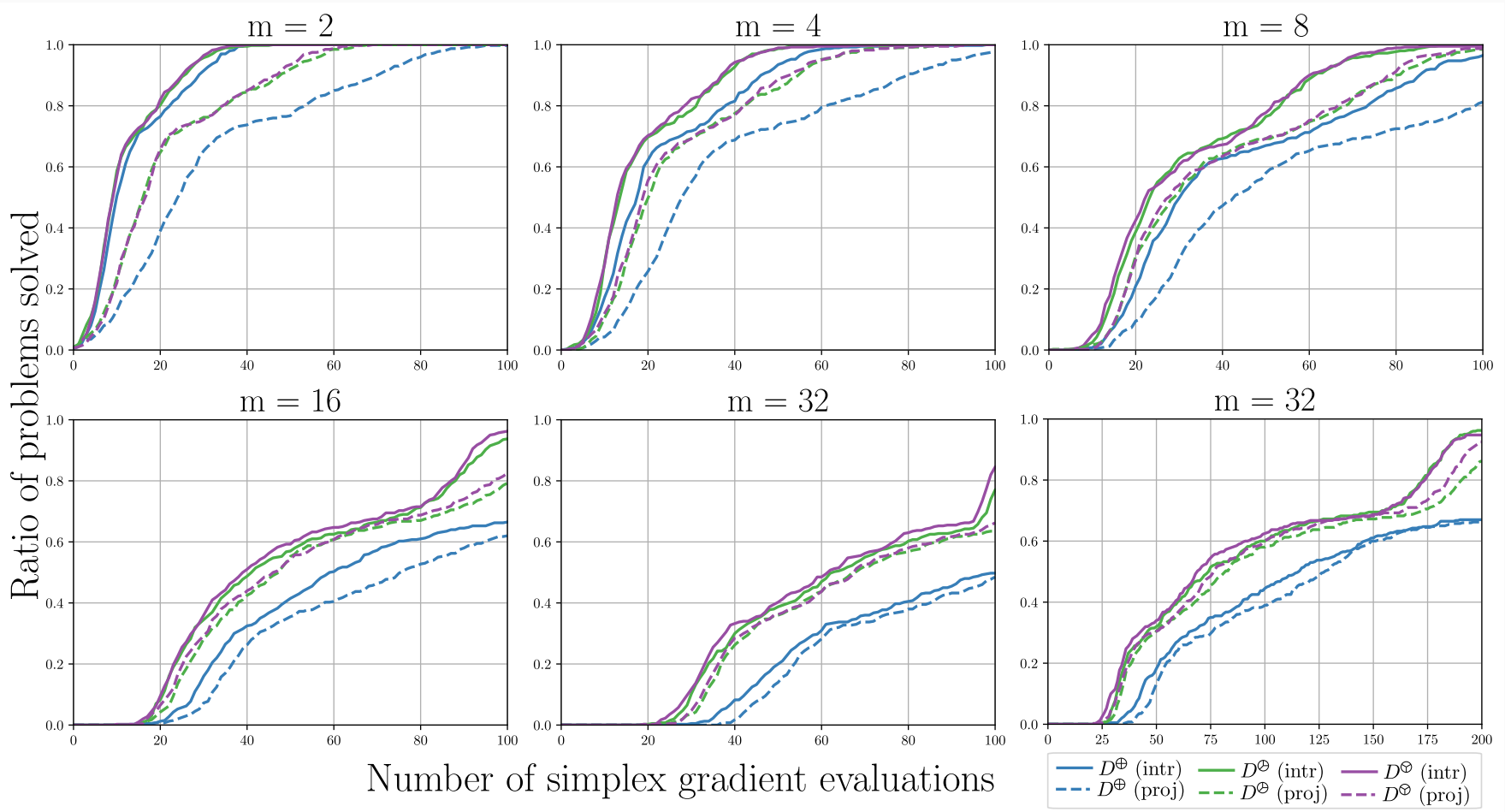
- 100 instances of each problem.
- Budget: $100(m+1)$ evaluations.
- Solved as soon as $\frac{f(x)-f_{\mathrm{low}}}{f(x_0)-f_{\mathrm{low}}} \leq 10^{-2}$.

Toy manifold problems



- Fixed $m = 4$. Varying $n - m$.
- Intrinsic PSSs outperform projected PSSs as $n - m$ grows.

Toy manifold problems



- Fixed $n - m = 4$. Varying m.
- Performance discrepancy fades as $\frac{n-m}{m} \to 0$.

Synchronization of rotations

$$\min_{R_1,R_2\in \mathrm{SO}(5)} \|R_1 - H_{12}R_2\|_F^2$$

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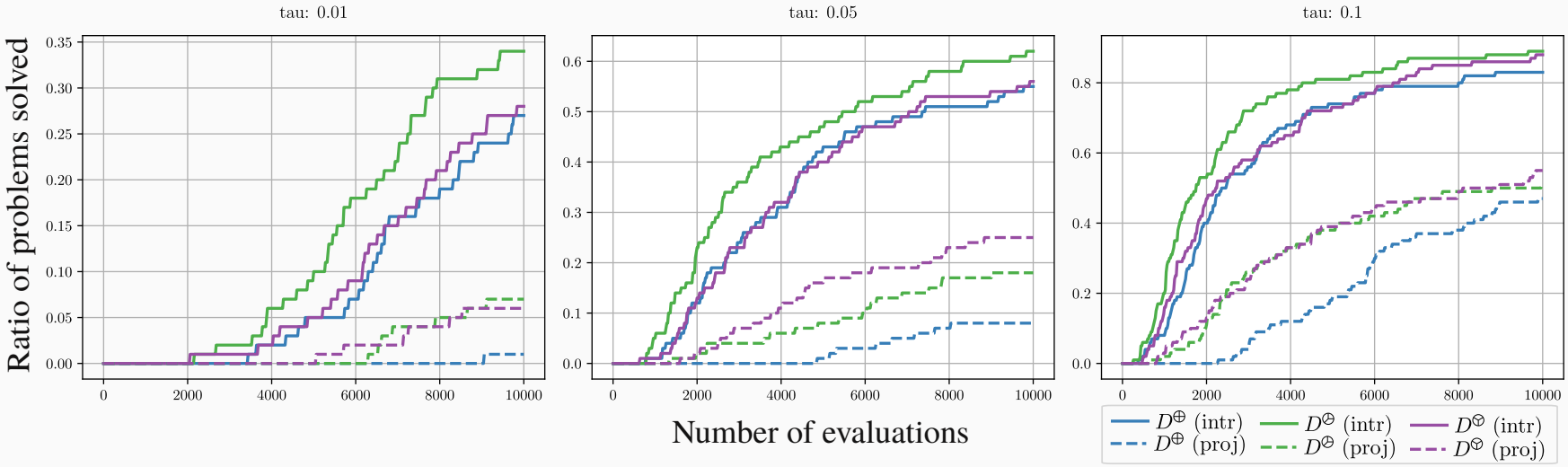
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- Budget: $500(m+1)$ evaluations.
- 100 problem instances (various $H_{12} \in \mathbb{R}^{5\times 5}$)
- Solved as soon as $\frac{f(x)-f_{\mathrm{low}}}{f(x_0)-f_{\mathrm{low}}} \leq \tau$, $\tau \in \{0.01, 0.05, 0.1\}$.

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Intrinsic PSSs (plain) still outperform projected PSSs.

Black-box adversarial attacks

Setup

- Network: Wide ResNet 28-10.
- Victims: 90 images from CIFAR10.



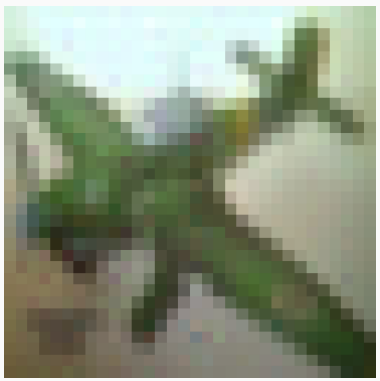
“truck”



“bird”



“frog”



“frog”

Black-box adversarial attacks

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- Perturbation: low-rank matrix $\delta = UV$ where $U, V^{\mathrm{T}} \in \mathbb{R}^{32 \times r}$.
- Problem: $\max_{\|\delta\|_2 \leq 0.5} f(\delta)$ with $f(\delta) = \mathrm{ce}(x + \delta)$



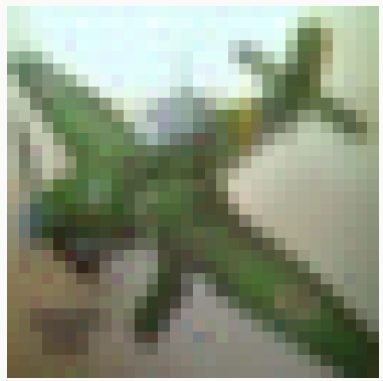
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- Problem: $\max_{\|\delta\|_2 \leq 0.5} f(\delta)$ with $f(\delta) = \mathrm{ce}(x + \delta)$
- Algorithms: Direct-search (DS), Probabilistic version (DS-PD), finite difference gradient descent (GFM)
- Budget: 30 720 queries.
- **Dataprofiles:** problems solved if $\frac{f(\delta)-f(\delta_*)}{f(\delta_0)-f(\delta_*)} < \frac{1}{10}$



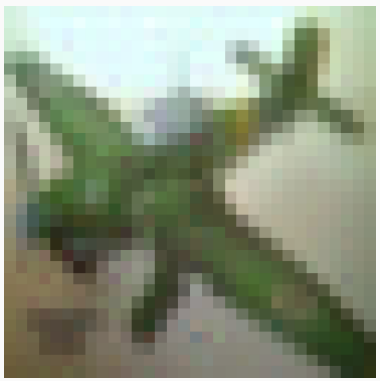
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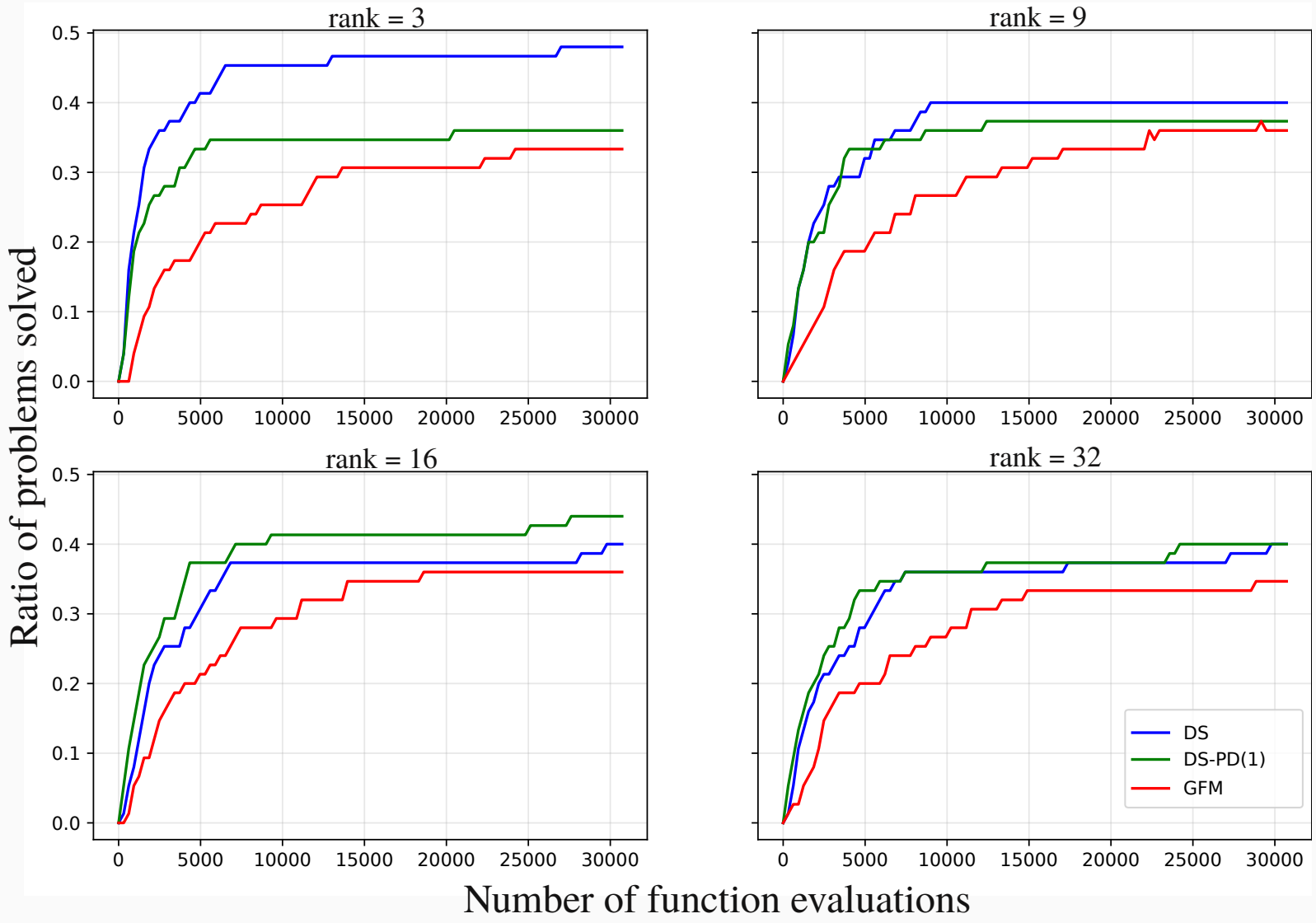


"frog"



"frog"

Black-box adversarial attacks



DS competes with finite-difference methods at equal budget

Conclusion

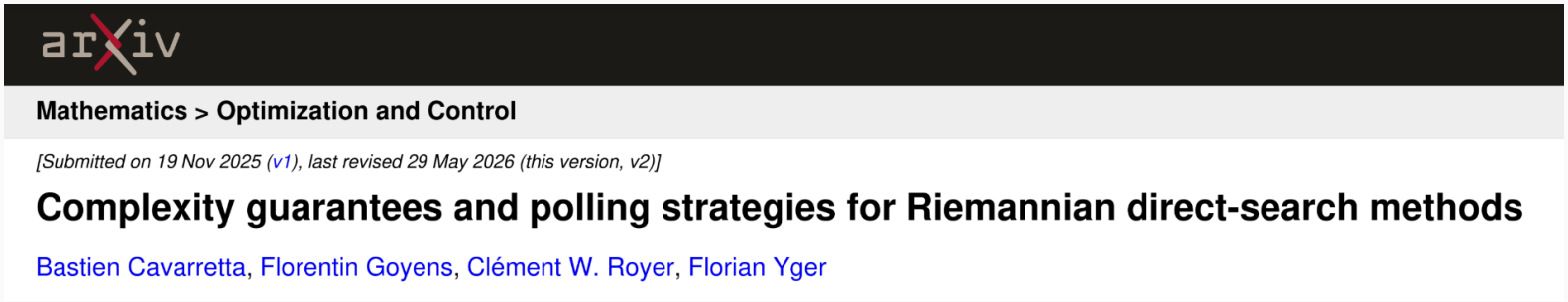
Our work

- Complexity analysis of Riemannian direct-search methods.
- Formalization of PSS for manifolds, advocate for intrinsics.

Direction

- More ambitious manifold application.
- Bound constraints $\longrightarrow$ Manifolds with corners ?

Vielen Dank


arXiv
Mathematics > Optimization and Control
[Submitted on 19 Nov 2025 (v1), last revised 29 May 2026 (this version, v2)]
Complexity guarantees and polling strategies for Riemannian direct-search methods
Bastien Cavarretta, Florentin Goyens, Clément W. Royer, Florian Yger